

Spring-2026

Saturday, 14th March

CSE-345 Computer Architecture and Assembly Language Midterm Exam

Total questions/marks: 40/40

Duration: 120 minutes

Name: _____

Student ERP: _____

Q1. What is the sum of your erp digits? **IM**

Q2. In Figure 1, if the branch instruction at memory location 0x20 is taken, then what instructions will be fetched from the instruction memory in the first 4 clock cycles? **IM**

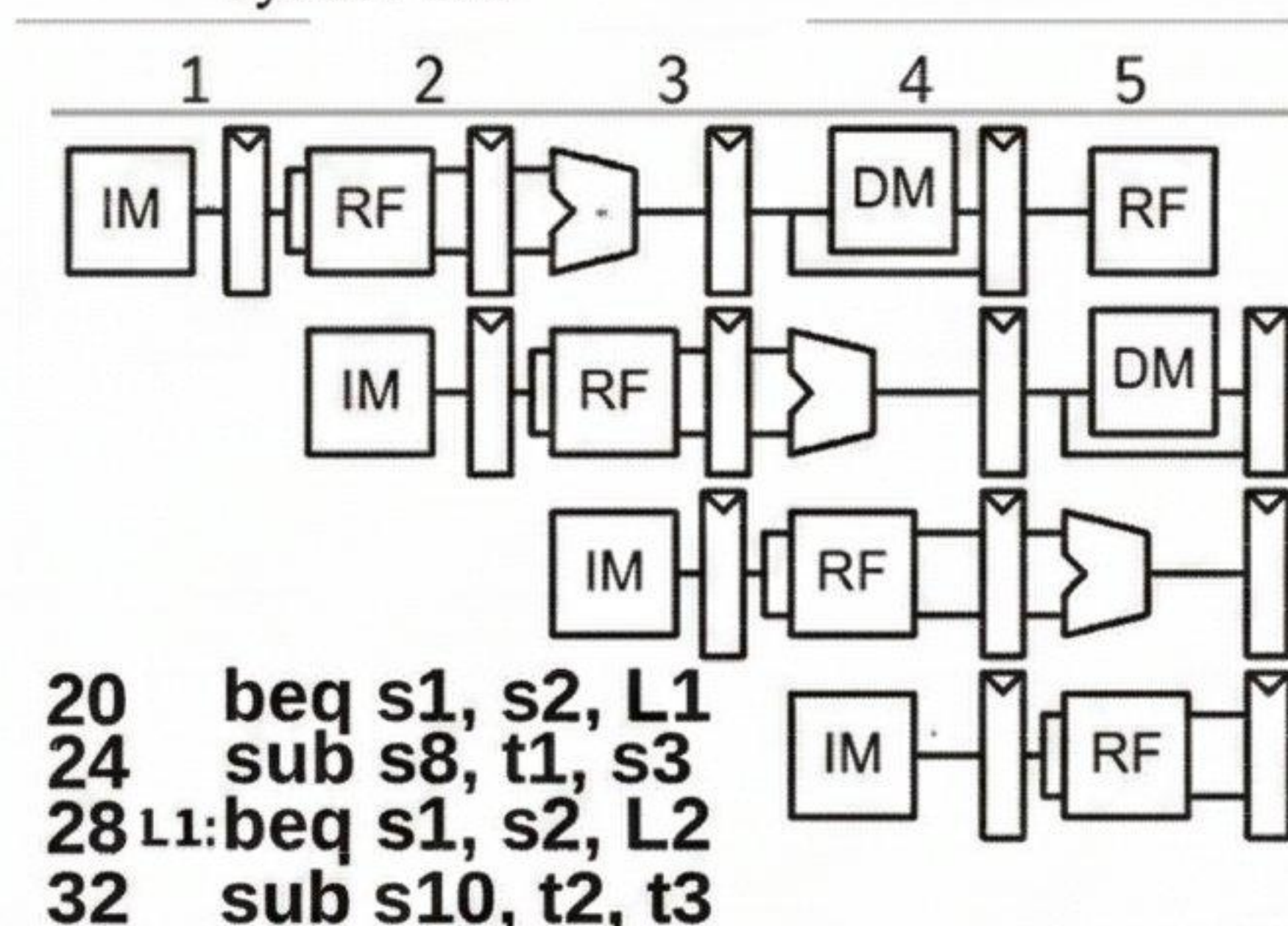


Figure 1

1. beq, sub, beq, sub
2. beq, sub, sub, sub
3. beq, sub, beq, beq
4. None of the rest

Q3. In Figure 1, if both the branch instructions are NOT taken, then what instructions will be fetched from the instruction memory in the first 4 clock cycles? **IM**

1. beq, sub, beq, sub
2. beq, sub, sub, beq
3. beq, sub, beq, beq
4. none of the rest

Q4. The role of the Extend Unit in the RISC-V architecture is to **IM**

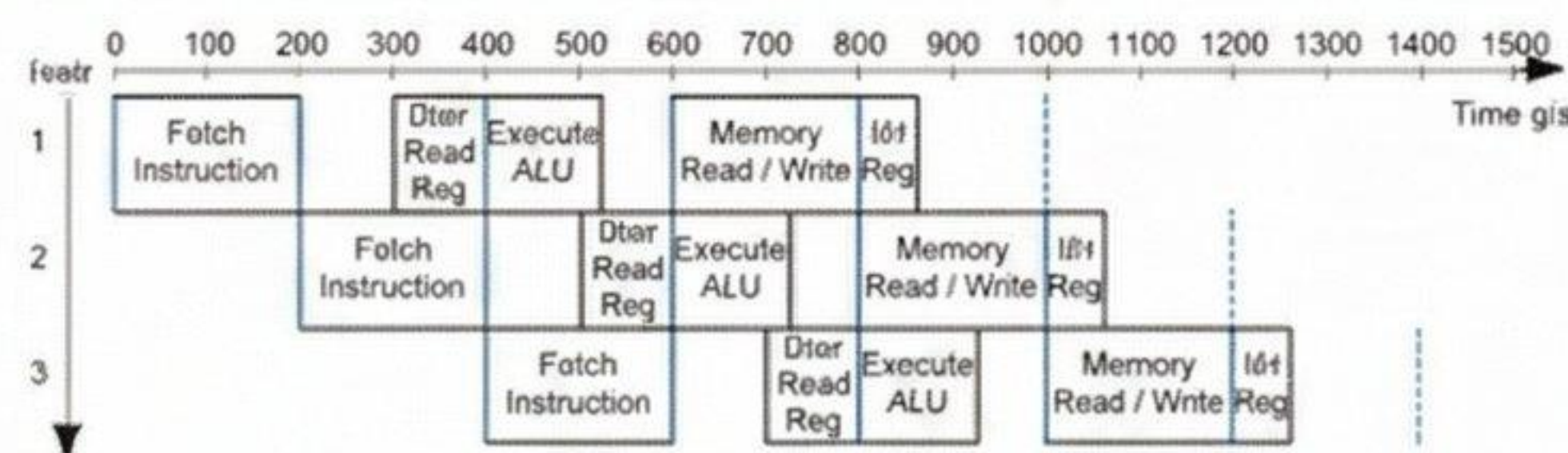
1. Multiply the input bit pattern by 2
2. Convert the input bit pattern into a 32-bit output bit pattern
3. Neither option 1 nor option 2
4. Both option 1 and option 2

Q4. Single instruction multiple data processing paradigm is an example of **IM**

1. Instruction-level parallelism
2. Data level parallelism

3. Thread-level parallelism
4. Memory level parallelism

Q5. Which of the stages will fail if $F_{clk} = 10\text{GHz}$? Ignore the t_{pcq} and t_{setup} **IM**



1. Fetch Only
2. Execute Only
3. Memory Only
4. All of These

Q6. This instruction format belongs to **IM**

31:25	24:20	24:15	14:12	11:7	6:0
funct7	rs2	rs1	funct3	rd	op

1. I-Type family
2. B-Type family
3. J-Type family
4. None of these

Q7. Contents of s1 after the execution of instruction `xori s1, zero, 0xFFFFFFFF` is **IM**

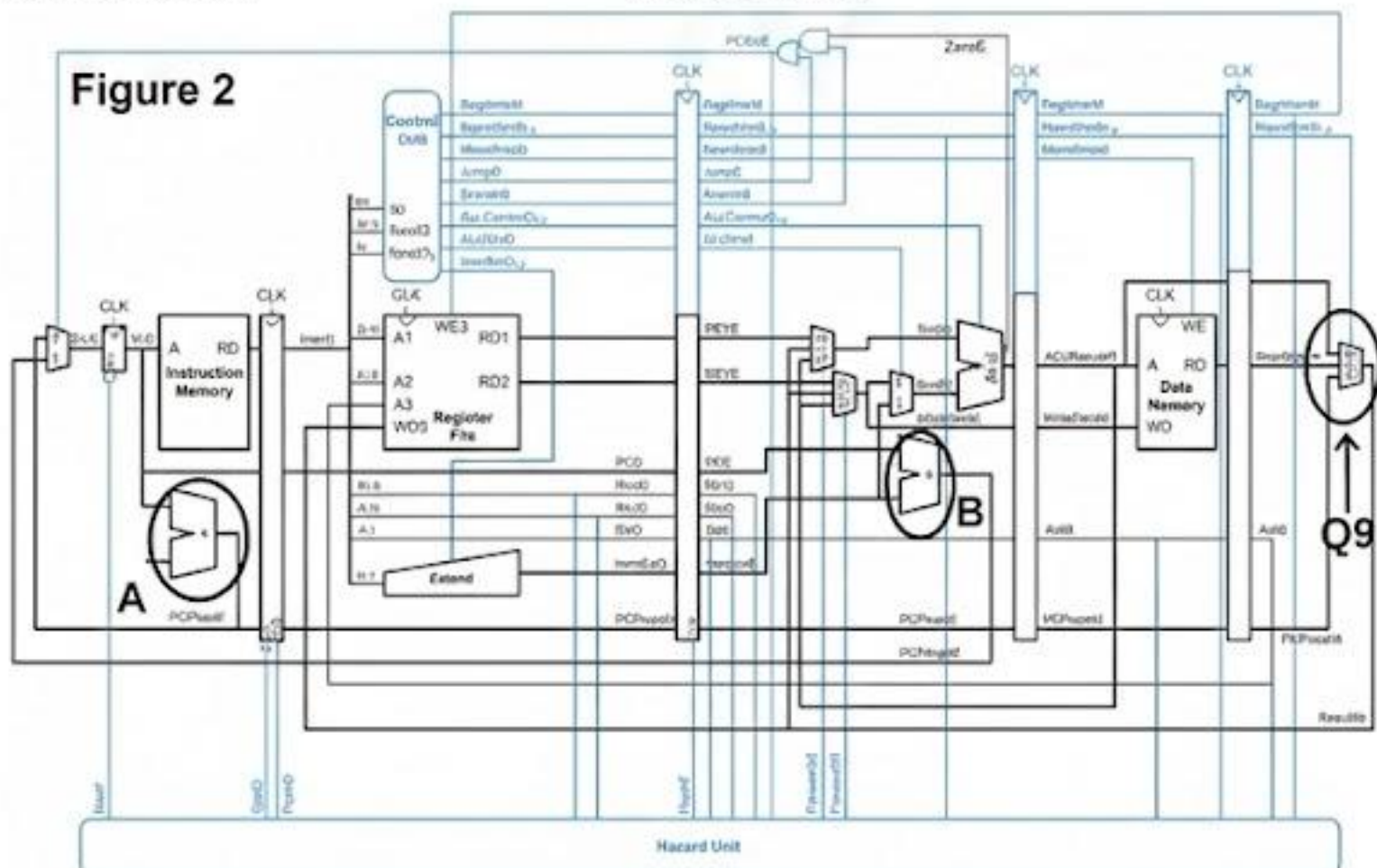
1. 0x1111 1111
2. 0xFFFF FFFF
3. 0x0000 0000
4. None of the above

Q8. In RISC-V vector assembly, if $VLEN=128$ bits, $LMUL=2$, $SEW=32$ bits, then the vectors at the application level can work with **IM**

1. 4 elements
2. 8 elements
3. 16 elements
4. None of these

Q9. In Figure 2, if the circled multiplexor in the WriteBack stage is pushed back into the Memory stage, then the clock period will **IM**

1. Increase



2. Decrease
3. Be unaffected
4. Be infinite

1 Mark

L7: lw x6, -4(x9)
sw x6, 8(x9)
or x4, x5, x6
beq x4, x4, L7

1. 0
2. 2
3. 4
4. None of the rest

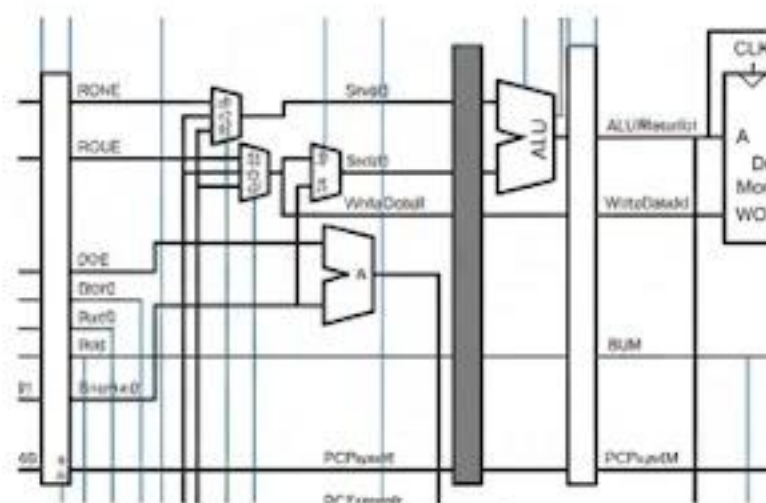


Figure 3

- Q10. If a new pipelined register is introduced as shown in Figure 3, then the clock period will be **IM**
1. Increase
 2. **Decrease**
 3. Be unaffected
 4. Be infinite
- Q11. In Single-Cycle implementation, determine the times the RAW hazard occurs in the following code snippet

- Q12. In execution of the **jal** instruction, as shown in Figure 2, which of the addres are used **IM**
1. Only A
 2. Only B
 3. Both A and B
 4. Both A and B, as well as ALU
- Q13. If the $T_c = 0.1 \times (\text{sum of erp digits})$ nanoseconds and $CPI = 1$ with 1 billion instructions, the program execution time will be **IM**
1. Greater than 1 second
 2. Less than 1 second
 3. Equal to 1 second
- Q14. A RISC-V vector program executes 6 instructions with the following profile in Table 1. What is the average CPI? **IM**
1. 4.5
 2. 5.0

3. 6.68

4. 6.25

Table 1

Instruction	Count	CPI
vsetvli	1	1
vle32.v	2	8
vfmacc.vv	1	4
vfadd.vv	2	8
vse32.v	2	8

Q15. Assume clock frequency = 2 GHz.
According to Table 1, what will be the total program execution time? 1M

1. 20.8 nanoseconds
2. 10 nanoseconds
3. 12.5 nanoseconds
4. 15.8 nanoseconds

Q16. If the hazard unit has a delay of $\tau_{hd}=10 \times$ the delay of squider my elighis gs. will be pipeline work correctly at $T=350$ ps? 1M

Component	Parameters	Delay, ps
Instruction unit (inst)	t_{inst}	30
Register group	t_{reg}	30
Memory access	t_{mem}	50
ALU and logic	t_{alu}	30
ALU	t_{alu}	200
Branch predictor (bpred)	t_{bpred}	10
Forward unit	t_{fwd}	30
Memory read	t_{mem}	200
Register file read	t_{reg}	100
Register file write	t_{reg}	40

1. Yes
2. No
3. Not enough information

Q17. In Figure 4, if $VLEN=256$ bits, $AVL=6$, then deccations. How many secondary registers will be accomplish the task in code 1M

1. B
2. 1
3. 2
4. 3

Q18. In Figure 4, if $VLEN=64$ bits, $AVL=6$, then and, 10], sil-1, 8 be replaced with

1. sil; 6, sil=6
2. sil; 8, 8
3. sil, 11, 8
4. None of the rest

```

4. .word 10, 20, 30, 10, 86, 06
y: .word 6, 10, 20, 20, 40, 06
z: .word 0, 0, 0, 0, 0, 8

vsetvli:0
vle32,v x2, z0, a02,
    a3, x2, xa
vfma2e vv a5 x4)
    lvn 12, x2, x6
    jll 00, 00, 7
    a06 a5, x0, 10
v(a0d x x2, (x5)
    add a2, a2, 10
vs32 av x2, v2, x1
vee32,v x6, (x5)
    add a2, a3, 10
    beqz a0, x0addine52
    ret
    
```

Figure 4

Q19. Consider a 5-stage RISC-V pipeline (IF, ID, EX, MEM, WB) with forwarding and hazard detection implemented, steel. There is no branch predictions, Autanse the following instruction sequences:

```

lw    x2, 0(x1)
add   x5, x2, x4
add   x6, x2, x4
beq   x3, x6, Label #taken
and   x5, x6, x3
    
```

Which of the following is the minimum number of stall on the reasonod for the correct execution of the instruction sequences, and vilget 1M

1. 1 stall cycle, due to a single load-use hazard.
2. 2 stall cycles, due to the load-use hazard and the branch dependency.
3. 3 stall cycles, reconsting for load-use hazard, branch resolution, and forwarding delay.
4. 3 stall cycles, considering both load-use and branch hazards along with data forwarding linitutions.

Q20. Which statement is true about a 1M

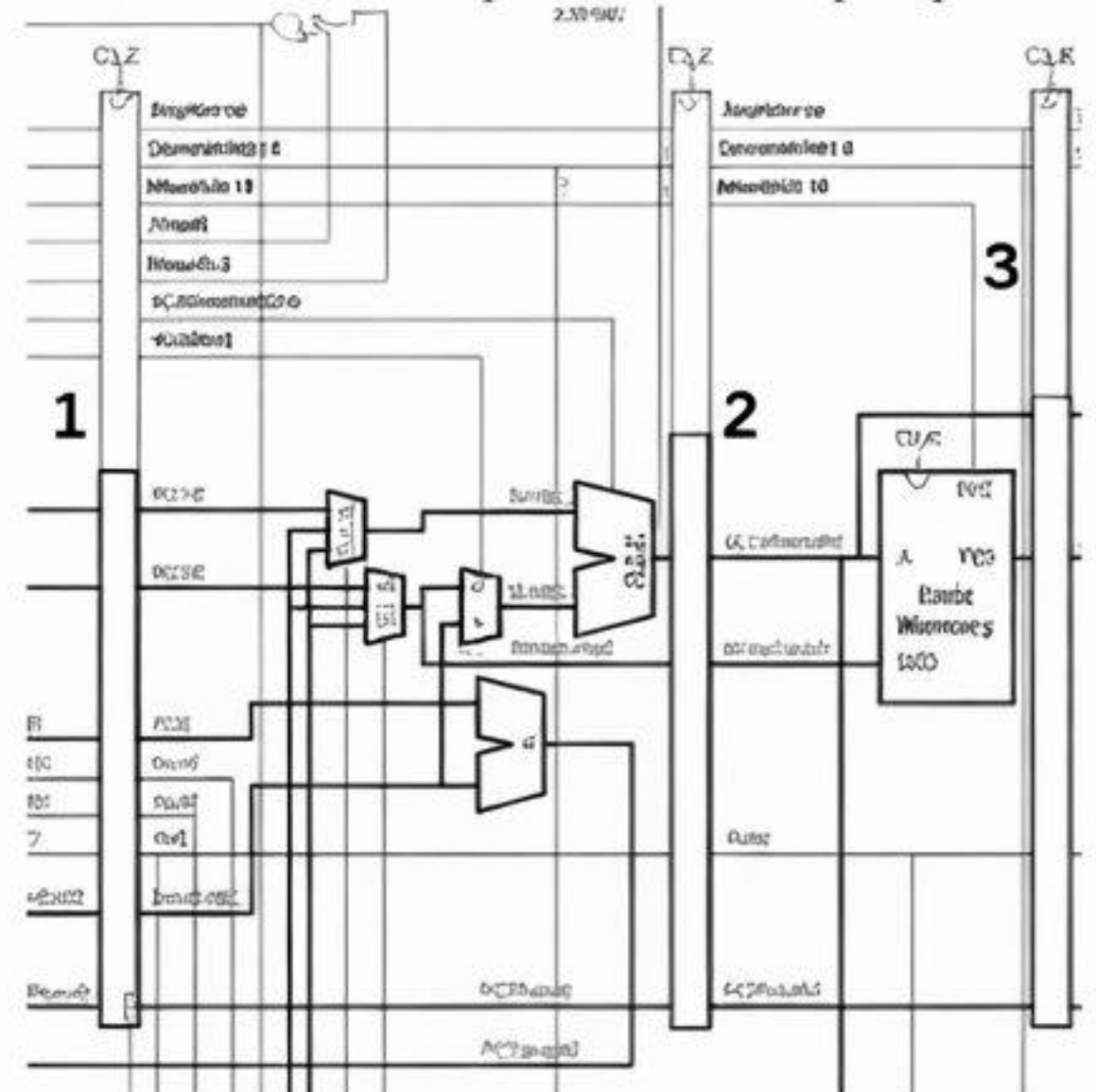
1. Cycles per instruction
2. Instructions per cycle
3. Cycles per three instructions
4. Instructions per two cycles

Q21. The following question is located al estd-address 0x0000002).

beq x1, x2, -16

1. 0x00400000
2. 0x00400010
3. 0x00400060
4. 0x00400064
- Q22. The following instruction is located at address 0x00400000
jal x0, 64
What is the Jump Target Address? IM
1. 0x00400040
2. 0x00400060
3. None of the rest
- Q23. In a hypothetical case, if there is an assumption of two hidden bits in branch size accommodation in the immediate field, then which of the following outputly defines the output of the extend unit IM
1. $PC + \text{SignExt}(\text{imm}_{16}, 1.001)$
2. $PC + \text{SignExt}(\text{imm}_{16}, 1.001)$
3. $PC + \text{SignExt}(\text{imm}_{16}, 20001)$
4. $PC + \text{SignExt}(\text{imm}_{16}, 20000)$
4. None of the rest
- Q24. The following instruction is at address 0x00400808:
auipc x5, 2
whose working principle is.
 $x5 \leftarrow \text{SignExt}(\text{imm}_{16}, 1'801) + 1'801 + PC$
What will be the value stored in x5? IM
1. $x5 = 0x00400020$
2. $x5 = 0x00402000$
3. $x5 = 0x00400200$
4. None of the rest
- Q25. Highlight the data and control signals' paths in Figure 2 for **auipc x5, 2**. Note: You will get IM only if you correctly highlight 4 paths.
1. Yes (For teacher use only)
2. No (For teacher use only)
- Q26. What is the range of go-to addresses in lw and instructions in RISC-V in terms of instruction IM:
1. -2048 to 2047
2. -512 to 511
3. -4066 to 4095
4. None of the rest
- Q27. Architecture and Microarchitecture are IM
1. Hardware and Instruction Set
2. Instruction Set and Software
3. Software and Hardware
4. Instruction Set and Hardware

- Q28. Which pipeline register, in Figure 5, has the maximum number of flip-flops? IM
1. 2
2. 3
3. 3
4. All have an equal number of flip-flops



- Q29. The register file _____ in single cycle tasce _____ in pipelined implementation IM
1. Negative edge triggered, positive edge triggered
2. Positive edge triggered, positive edge triggered
3. Negative edge triggered, negative edge triggered
4. None of the rest
- Q30. A program has 20% branch instructions. The sealar pipelined procesr or has a misprediction sale of 10% and a penalty of 2 cycles. Meal CPI = 1. What is the effective
1. 0.8, IM
2. 1.01
3. 1.10
4. 1.20
- Q31. A 2+2 matrix is stored in row-major order in memory starting at base address x1 = 0x2000

Original Matrix A: Memory Layout:

1	2
3	4

Address 0x2000 = 1 (A[0][0])
Address 0x2004 = 2 (A[0][1])
Address 0x2008 = 3 (A[1][0])
Address 0x200C = 4 (A[1][1])

Address 0x200C = 4 (A[[1[1]]])

To transpose this matrix, the following code is executed:

x1 = 0x2000 (base address)

Each element is 4 bytes (32-bit integer)

vsetivli x0, 4, e32, m1

vluxei32.v v2, (x1), v1

Which index vector v1 (in byte offsets) correctly gathers elements to produce the transposed matrix {1, 3, 2, 4} in v2? **1M**

1. v1 = {0, 4, 8, 12}
2. v1 = {0, 8, 4, 12}
3. v1 = {4, 0, 12, 8}
4. v1 = {12, 8, 4, 0}

Q32. In jal instruction, the jump address is _____, and the link address is _____ **1M**

1. PC+4, JTA
2. JTA, PC+4
3. JTA/2, PC+4
4. None of the rest

Q33. Program execution time in pipelined implementation reduces due to **1M**

1. Increased T_c , Decreased CPI
2. Increased CPI, Decreased T_c
3. Increased CPI, Increased T_c
4. Decreased CPI, Decreased T_c

Q34. There are 4 operational stages in RISC-V pipelined implementation **1M**

1. True
2. False

Q35. The following instructions respectively belong to **1M**

lw x6, -4(x9)

sw x6, 8(x9)

or x4, x5, x6

beq x4, x4, L7

1. S, I, B, R
2. I, S, R, B
3. I, S, B, R
4. S, I, R, B

Q36. What family of instructions do these hex codes represent? **1M**

0064A423

FFC4A303

FE420AE3

0062E233

1. S, I, B, R
2. I, S, B, R
3. S, I, R, B
4. I, S, R, B

Q37. Which operational stage does the control signal RegWriteM belong to **1M**

1. Writeback stage
2. Execute stage
3. Decode Stage
4. None of the rest

Q38. For the code

addi s0, zero, 6

addi s1, zero, 0

LOOP:

beq s0, zero, DONE

lw s2, 0(s1)

lw s3, 16(s1)

lw s4, 32(s1)

lw s5, 48(s1)

addi s0, s0, -1

j LOOP

DONE:

What will be stored in the immediate field of the beq instruction? **1M**

1. 28
2. ?
3. 14
4. None of the rest

Q39. RISC-V is **1M**

1. IP of Intel
2. Open-source
3. IP of AMD
4. None of the rest

Q40. RV32I is **1M**

1. Vector Extension
2. Base ISA
3. Integer Extension
4. None of the rest

*****The End*****